

CSE213 CRA Report

Autumn 2025

Topic: Simulating operations of a Large scale Mango-Garden estate		
ID: 2432008	Name: Raiyan Bin Rais	User1: Owner
		User2: General Manager
ID:2430825	Name: Mohammad Zaid Iqbal Fahad	User3: Accountant
		User4: Field Supervisor
ID: 2430794	Name: Rashedur Rahman Arafat	User5: Warehouse Manager
		User6: Transport Manager
ID: 2430936	Name: Juyairia Benta Alam	User7: Garden Worker
		User8: Customer

P.T.O

Common Processes:

Process-1: Login			
			event type
Workflow :	event-1	Open Login page	UIE
	event-2	Enter username and password	UID
	event-3	Validate the inputs (required, format)	VL
	event-4	Check if the username-password combination is correct	VR
	event-5	If correct, redirect user to dashboard, otherwise show “invalid”	OP

Process-2: Log Out			
			event type
Workflow :	event-1	Click on Log Out button	UIE
	event-2	Invalidate session	DP
	event-3	Redirect user to the login page	OP

Process-3: Update Profile Info			
			event type
Workflow :	event-1	Open “Profile”	UIE
	event-2	Edit profile fields	UID
	event-3	Validate fields	VL

	event-4	Save changes	DP
	event-5	Show confirmation	OP

Workflow of user specific goals:

User-1 name: Owner			
User-1, Goal-1	Description of Goal-1: View Task Summary		event type
Workflow :	event-1	Open Task summary page	UIE
	event-2	Fetch task reports from General Manager	DP
	event-3	Display reports	OP
User-1, Goal-2	Description of Goal-2: View Inventory Summary		
	event-1	Open inventory summary page	UIE
	event-2	Fetch inventory summary	DP
	event-3	Display report	OP
User-1, Goal-3	Description of Goal-3: View Sales/Orders summary		
	event-1	Open sales summary page	UIE
	event-2	Fetch sales summary	DP
	event-3	Display report	OP
User-1, Goal-4	Goal-4: View all employees		
	event-1	Fetch all employees	DP
	event-2	Display employee list	OP
User-1, Goal-5	Goal-5: Add new Employees		

	event-2	Enter employee details	UID
	event-3	Validate input	VL
	event-4	Save to database	DP
	event-5	Display confirmation	OP
	event-6	Select employee to delete	UID
	event-7	Delete the employee and update database	DP
User-1, Goal-6	Goal-6: Remove/Fire existing employees		
	event-1	Load and display employee list	DP
	event-2	Select employee to remove/fire	UID
	event-3	Click “remove”	UIE
	event-4	Update database and display confirmation	DP
User-1, Goal-7	Description of Goal-7: Approve Special Orders		
Workflow :	event-1	Fetch pending special orders	DP
	event-2	Approve/Reject	UIE
	event-3	Save decision	DP
	event-4	Notify Customer	OP
User-1, Goal-8	Description of Goal-8: Approve Large Expenses		
Workflow :	event-1	Fetch pending large expenses	DP
	event-2	Review details	OP
	event-3	Approve/Reject	UIE
	...	Save decision	DP
	event-n	Notify requester	OP

User-2 name: General Manager			
User-2, Goal-1	Description of Goal-1: View all pending budget requests		Event type
	event-1	Fetch budget requests from Accountant	DP
	event-2	Display the requests in a tableview	OP
User-2, Goal-2	Description of Goal-2: Approve/reject budget requests		event type
Workflow :	event-1	Show pending budget approval requests	OP
	event-2	Review request details (requester, department, amount, justification)	OP
	event-3	Select action (Approve/Reject)	UIE
	event-4	If approved, update budget allocation accordingly and notify all relevant users	DP
User-2, Goal-3	Description of Goal-3: Create and assign Tasks to Supervisors		
Workflow :	event-1	Open task assignment page	UIE
	event-2	Select field supervisor	UID
	event-3	Enter task details & steps	UID
	event-4	Save assignment	DP
	event-5	Display confirmation	OP
	event-6	Notify the supervisor to whom the task has been assigned	OP
User-2, Goal-4	Description of Goal-4: Track Task Completion		
Workflow	event-1	Fetch supervisor reports	DP

:			
	event-2	Display summary	OP
User-2, Goal-5	Description of Goal-5: Check Inventory		
Workflow :	event-1	Fetch inventory from Warehouse Manager	DP
	event-2	Display inventory summary	OP
User-2, Goal-6	Description of Goal-6: Generate Periodical Task Summary		
Workflow :	event-1	Select time period (starting, ending)	UIE
	event-2	Fetch all tasks for that period (date validation)	DP
	event-2	Generate reportreport	DP
User-2, Goal-7	Description of Goal-7: Set/modify employee salary rates		
	event-1	Fetch employee types & salary rates	DP
	event-2	Modify (pay interval/rate)	UID
	event-3	Save changes	DP
User-2, Goal-8	Description of Goal-8: Calculate worker salary		
Workflow :	event-1	Access Payroll module	UIE
	event-2	Select worker and payroll period (e.g., current month)	UIE
	event-3	Verify attendance records and work records of each worker	VR
	event-4	Calculate gross wages (base + overtime)	DP
	event-5	Apply deductions (tax, advances)	DP

	event-6	Review calculated net wages	OP
	event-7	Save payroll calculation to database	DP

User-3 name: Accountant			
User-3, Goal-1	Description of Goal-1: Record daily income or expense transactions in the ledger.		event type
Workflow :	event-1	Enter details (Amount, Type, Category, Date, Description)	UIE
	event-2	Validate fields	UID
	event-3	Save entry	UID
User-3, Goal-2	Description of Goal-2: View/edit transaction records		event type
Workflow :	event-1	Fetch transaction records	DP
	event-2	Show all transaction records in TableView	UIE
	event-3	Select transaction record to edit	UIE
	event-5	Edit the fields accordingly	DP
	event-6	Save changes and update	OP
User-3, Goal-3	Description of Goal-3: View Employee salary history		
Workflow :	event-1	Fetch salary records from file	DP
	event-2	Display them in a Table	OP
User-3, Goal-4	Description of Goal-4: Disburse salaries to workers		
Workflow :	event-1	Select payroll period (e.g., current month) to pay	UIE
	event-2	System loads the previously calculated payroll for the workers	DP
	event-3	Choose payment method(s): Bank Transfer /	UID

		Mobile Wallet / Cash	
	event-4	Execute payment	DP
	event-5	Save to file	DP
User-3, Goal-5	Description of Goal-5: Create & submit budget approval request to the General Manager		
Workflow :	event-1	Access "Budget Requests" module	UIE
	event-2	Select budget type and enter other details (amount, purpose, etc.)	UID
	event-3	Validate fields (required entries, positive amounts, valid dates); highlight errors	VL
	event-4	Click on "Submit Request" to submit	UIE
User-3, Goal-6	Description of Goal-6: Create and submit large expense approval request to the Owner for approval		
Workflow :	event-1	Access "Large Expenses" module	UIE
	event-2	Input large expense details accordingly	UID
	event-3	System validates all fields and highlights errors	VL
	event-4	If all okay, user clicks "Submit" and the request is submitted	UIE
User-3, Goal-7	Description of Goal-7: Process Requests for Tools/Resources		
Workflow :	event-1	Fetch requests from Field Supervisor	DP
	event-2	Review	OP
	event-3	Approve/Reject	UIE
	event-4	If approved, generate budget request	DP
	event-5	Submit budget request to General Manager	OP

User-3, Goal-8	Description of Goal-8: Process Requests for Vehicle Maintenance		
Workflow :	event-1	Fetch requests from Transport Manager	DP
	event-2	Review	OP
	event-3	Approve/Reject	UIE
	event-4	If approved, generate budget request	DP
	event-5	Submit budget request to General Manager	OP

User-4 name: Field Supervisor			
User-4, Goal-1	Description of Goal-1: Assign Daily Tasks to Workers		event type
Workflow :	event-1	Fetch assigned tasks from manager	UID
	event-2	Select task to assign	UID
	event-3	Select workers to assign	UID
	event-4	Save	DP
User-4, Goal-2	Description of Goal-2: Record Worker Attendance		
Workflow :	event-1	Mark attendance	UID
	event-2	Save attendance record	DP
User-4, Goal-3	Description of Goal-3: Monitor Task Completion		
Workflow :	event-1	Fetch task reports from workers	DP
	event-2	Load the tasks and progress	DP
User-4, Goal-4	Description of Goal-4: Provide Instructions/Feedback for Field Problems/Issues		
Workflow :	event-1	Fetch issue reports from workers	DP
	event-2	Enter instructions/feedback	UID
	event-3	Save instructions/feedback	DP
User-4, Goal-5	Description of Goal-5: Request Tools/Resources		
Workflow	event-1	Enter request	UID

:			
	event-2	save	DP
	event-3	notify Accountant	OP
User-4, Goal-6	Description of Goal-6: Generate Daily Field Summary		
Workflow :	event-1	Select date	UID
	event-2	Click “Generate”	UIE
	event-3	System generates and shows summary	DP
	event-2	Save record	OP
User-4, Goal-7	Description of Goal-7: Suggest Improvements		
Workflow :	event-1	Enter suggestions	UID
	event-2	save	DP
	event-3	notify Manager	OP
User-4, Goal-8	Description of Goal-8: Monitor Field Resource Inventory (tools, pesticides etc.) Usage		
Workflow :	event-1	Load up current field resource inventory levels	OP
	event-2	Select an item to log usage	UID
	event-3	Submit usage log	DP
	event-4	Update database and show updated inventory	OP

User-5 name: Warehouse Manager			
User-5, Goal-1	Description of Goal-1: Record harvested mango batch		event type
Workflow :	event-1	• Enter BatchID • Enter HarvestDate • Enter field number	UID
	event-2	Save batch information in database	DP
	event-3	Confirm message that the mango batch has been successfully recorded	OP
User-5, Goal-2	Description of Goal-2: Mango Inventory		
Workflow :	event-1	Display details(BatchID,HarvestDate,Quantity,FieldNumber, MangoType,MangoSpoile)	UID
User-5, Goal-3	Description of Goal-3 : Update inventory		
Workflow :	event-1	Select the BatchID	UID
	event-2	Select the add or remove mango	UID
	event-3	Enter the mangoQuantity	DP
	event-4	ConfirmMessage that inventory records have been successfully updated	OP
User-5, Goal-4	Description of Goal-4: Track spoilage Monitor and record spoiled or damaged mangoes		
Workflow :	event-1	Enter the number or weight of spoiled/damaged mangoes	UID

	event-2	Reduce inventory by the spoiled quantity	DP
	event-3	Store updated data for tracking losses and analysis	DP
User-5, Goal-5	Description of Goal-5: Approve order and prepare Approve customer orders and prepare them for delivery		
Workflow :	event-1	Choose the customer order ready for dispatch	UIE
	event-2	Record dispatch approval and update order delivery status	DP
			OP
User-5, Goal-6	Description of Goal-5: Send prepared order to Transport Manager for delivery Forward ready orders to Transport Manager for delivery		
	event-1	View all orders that are ready for dispatch	OP
	event-2	Send the delivery request to Transport Manager	OP
	event-3	Update order status in the system	DP
User-5, Goal-7	Description of Goal-7: Generate inventory summary Create a summary report of current inventory		
Workflow :	event-1	● Gather total stock information ● Include available, spoiled, and dispatched quantities	DP
	event-2	● Generate the summary report ● Save it with a timestamp	DP
User-5, Goal-8	Description of Goal-8: Request harvest Monitor inventory and request harvest as needed.		

Workflow	event-1	Review current stock levels	OP
	event-2	Send harvest request to Field Supervisor	OP

User-6 name: Transport Manager

User-6, Goal-1	Description of Goal-1: Assign Vehicles		event type
Workflow :	event-1	• Receive delivery order from Warehouse Manager • Check order details	UIE
	event-2	• Load vehicles list • Choose a suitable vehicle • Check capacity • Check availability • Assign order to the vehicle	DP UID VL VL DP
	event-3	• Send confirmation message • Confirm vehicle assigned successfully	OP
User-6, Goal-2	Description of Goal-2: Track Deliveries		
	Check and monitor the live delivery status		
Workflow :	event-1	• Enter delivery order ID	UID
	event-2	• Fetch delivery status	DP
User-6, Goal-3	Description of Goal-3: Record Delivery Completed		
	Mark a delivery as completed and save the final details		
Workflow :	event-1	• Open active deliveries list	UIE UID

		Choose the completed delivery order	
	event-2	Add completed date	UID
	event-3	Update order status	DP
User-6, Goal-4	Description of Goal-4: Schedule Maintenance Plan regular vehicle maintenance		
Workflow :	event-1	Load vehicles list	DP
	event-2	Choose the vehicle needing inspection or maintenance	UID
	event-3	Add maintenance date Add relevant notes	UID
	event-4	Store maintenance schedule Keep data for future reminders	DP
User-6, Goal-5	Description of Goal-5: Assign Drivers Assign drivers to vehicles		
Workflow :	event-1	Load vehicles and drivers list	OP
	event-2	Pick an available and eligible driver.	UIE
	event-2	Save driver assignment details in the system	DP
User-6, Goal-6	Description of Goal-6: Request Vehicle Servicing Budget Send request for vehicle servicing budget to the Accountant		
Workflow :	event-1	Load vehicle list / schedule maintenance list	OP
	event-2	Select vehicle that needs servicing	UIE
	event-3	Create request for servicing budget	UID
	event-3	Submit request to Accountant	DP

	event-4	Notify accountant	OP
User-6, Goal-7	Description of Goal-7: View Delivery Logs Check past and ongoing deliveries		
Workflow :	event-1	Get completed and ongoing delivery records	DP
	event-2	Display details: time, vehicle used, driver info	OP
User-6, Goal-8	Description of Goal-8: Generate Delivery Summary Create a report summarizing deliveries		
Workflow :	event-1	Gather key transport info (total deliveries, distance, completion rate)	DP
	event-2	Show summarized report in an organized format	OP

User-7 name: Garden Worker			
User-7, Goal-1	Description of Goal-1: View Assigned Tasks		event type
Workflow :	event-1	Get the list of daily field assignments allocated to the worker from the Field Supervisor.	DP
	event-2	Show detailed task information including location, crop type, and deadlines.	OP
User-7, Goal-2	Description of Goal-2: Mark Task Complete		
Workflow :	event-1	Choose the completed task from the assigned task list.	UIE
	event-2	Update the task status to indicate completion.	UIE
	event-3	Store the completion record in the system.	DP
User-7, Goal-3	Description of Goal-3: Record Daily Harvest Details		
Workflow :	event-1	Input the total harvested mango quantity along with quality remarks or grades.	UID
	event-2	Save the harvest record and send to Warehouse manager for updating inventory	DP
User-7, Goal-4	Description of Goal-4: Request Tools		
Workflow :	event-1	Specify the name and quantity of tools required for field operations.	UID
	event-2	Record the tool request in the system for field supervisor review and approval.	DP
User-7, Goal-5	Description of Goal-5: Report Field Issues		

Workflow :	event-1	Describe any problem noticed in the field.	UID
	event-2	Save the issue record and send notification to Field Supervisor	DP
User-7, Goal-6	Description of Goal-6: View Attendance report		
Workflow :	event-1	Retrieve the attendance by the supervisor.	DP
	event-2	Show attendance	OP
User-7, Goal-7	Description of Goal-7: View Salary history		
Workflow :	event-1	Access own record of salary month wise	UID
	event-2	Store the daily summary for attendance and performance evaluation.	DP
User-7, Goal-8	Description of Goal-8: View Work History		
Workflow :	event-1	Access own record of work completion (marked by date and task)	OP
	event-2	Show the work history in a TableView	UIE

User-8 name: Customer			
User-8, Goal-1	Description of Goal-1: Place an order Allow the customer to create and submit a mango order.		event type
Workflow :	event-1	Customer views the order page (mango types, in stock/out of stock)	UID
	event-2	Customer selects type, inputs quantity, delivery address, and other required order information	UID
	event-3	Ensure all details are accurate and complete, including quantity and address.	VL
	event-4	Save the validated order in the system for processing	DP
	event -5	If order amount cross 50,000 tk add order to special order for owner approval	VL
	event-6	Create an order and notify the Warehouse Manager	DP
User-8, Goal-2	Description of Goal-2: Track an order Allow the customer to check the status of their placed order		
Workflow :	event-1	Customer inputs the unique order identification number to start tracking.	UID
	event-2	System fetches the stored order data from the database using the provided Order ID.	DP
	event-3	Show the customer the current status, location, and delivery progress of their mango order	OP
User-8, Goal-3	Description of Goal-3: Cancel order Allow the customer to cancel a placed order if conditions permit.		
Workflow :	event-1	Customer enters the order number	UID

	event-2	System checks if the order can be cancelled based on status and time limits.	VL
	event-3	Record the cancellation request and update the order status in the system	DP
	event-4	Send confirmation to the user that the mango order has been successfully cancelled.	OP
	event-5	Notify warehouse manager that the order has been cancelled	OP
User-8, Goal-4	Description of Goal-4: View order history Allow the customer to check all previous mango orders.		
Workflow :	event-1	Fetch all past mango orders from the database linked to the customer's account.	DP
	event-2	Show the full order list with dates, order IDs, and statuses for the customer to review	OP
User-8, Goal-5	Description of Goal-5: Make payments Allow the customer to pay for their mango order securely.		
Workflow :	event-1	Load up the orders list. Select order for payment	UIE
	event-2	Check payment info, amount, and whether the order is eligible for payment.	VL
	event-2	Save the payment securely and update the order status to "Paid."	DP
	event-3	Provide the customer with a confirmation message and payment receipt.	OP
User-8, Goal-6	Description of Goal-6: Request invoice		
Workflow :	event-1	Select order from order list	UID

	event-2	Create a detailed invoice including order summary, pricing, and payment information. Store the invoice in the system.	DP
	event-3	Display or allow the customer to download the generated invoice.	OP
User-8, Goal-7	Description of Goal-7: Provide feedback		
Workflow :	event-1	Allow the customer to write and submit feedback regarding the mango order or service.	UID
	event-2	Save the submitted feedback in the system for quality tracking and analysis.	DP
	event-3	Send an alert to the administrator that new customer feedback has been received.	OP
User-8, Goal-8	Description of Goal-8: Contact support		
Workflow :	event-1	The customer sends a support request describing the issue or question related to their order	UID
	event-2	The system confirms that the support request has been received and informs the customer that the support team will respond soon.	OP